

Prompt Confusion Phase 04

Combined Stages 1 & 2 Report

April 10, 2026 — conflict detection, arbitration probing, causal intervention, confound analysis, and attention follow-up.

BOTTOM LINE

The model knows when its instructions conflict (92% probe accuracy). We can causally steer which side it favors. But the arbitration signal is not a clean “source-selection mechanism” — it is dominated by the *type* of conflict and contextual pressure, not an abstract resolution circuit. That is itself informative: the model’s conflict resolution is family-specific and pressure-responsive, not generic.

Plain-language summary

We gave Qwen 288 synthetic trading prompts — half with internally consistent instructions, half where SETTINGS and STRATEGY deliberately contradict each other. Here is what we learned across two rounds of analysis:

1. **The model builds a clear “conflicted” representation.** A linear probe on the residual stream detects conflict at 92% accuracy (layer 36). This is a strong, clean finding.
2. **The model can be steered.** Injecting the SETTINGS direction into the residual stream at layer 24 moves behavior in the expected direction. `strategy_push` shifts the model 9 percentage points toward strategy-following; `setting_push` shifts it about 3 points toward setting-following.
3. **But “which side wins” is mostly determined by the type of conflict.** A simple classifier using just the conflict family and pressure bucket predicts the winner at 80% — *better* than the residual probe (71%). The model isn’t using one general-purpose arbitration circuit; it is resolving different kinds of conflicts differently. That is a real finding about conflict resolution, just not the one we initially expected.
4. **Attention is not yet informative.** Section-aggregated attention barely beats chance as an arbitration decoder (best: 59%).

Run Overview

Behavior

16474bce

288 / 288 valid structured outputs.

Conflict Probe

62fc89fd

Aligned vs. conflict, 92% peak.

Arbitration Probe

6924c2b3

Conflict-only, 71% peak.

Causal Rerun

Stage 2

Harness fixed. Intervention active.

Confound Check

Stage 2

Family + pressure = 80% baseline.

Attention

Stage 2

Weak decoder, best 59%.

1. Conflict Detection Is Strong

Can we read *whether a conflict exists* from the residual stream? Yes — cleanly and reliably.

Layer	Balanced Accuracy	Std Dev	Selectivity
0	0.784	0.038	0.310
20	0.788	0.024	0.281
24	0.864	0.037	0.399
28	0.861	0.021	0.368
36	0.920	0.042	0.417
44	0.837	0.072	0.354

This is a Level-2 representational result. The model builds an increasingly clear internal representation of “these instructions contradict each other,” peaking at layer 36.

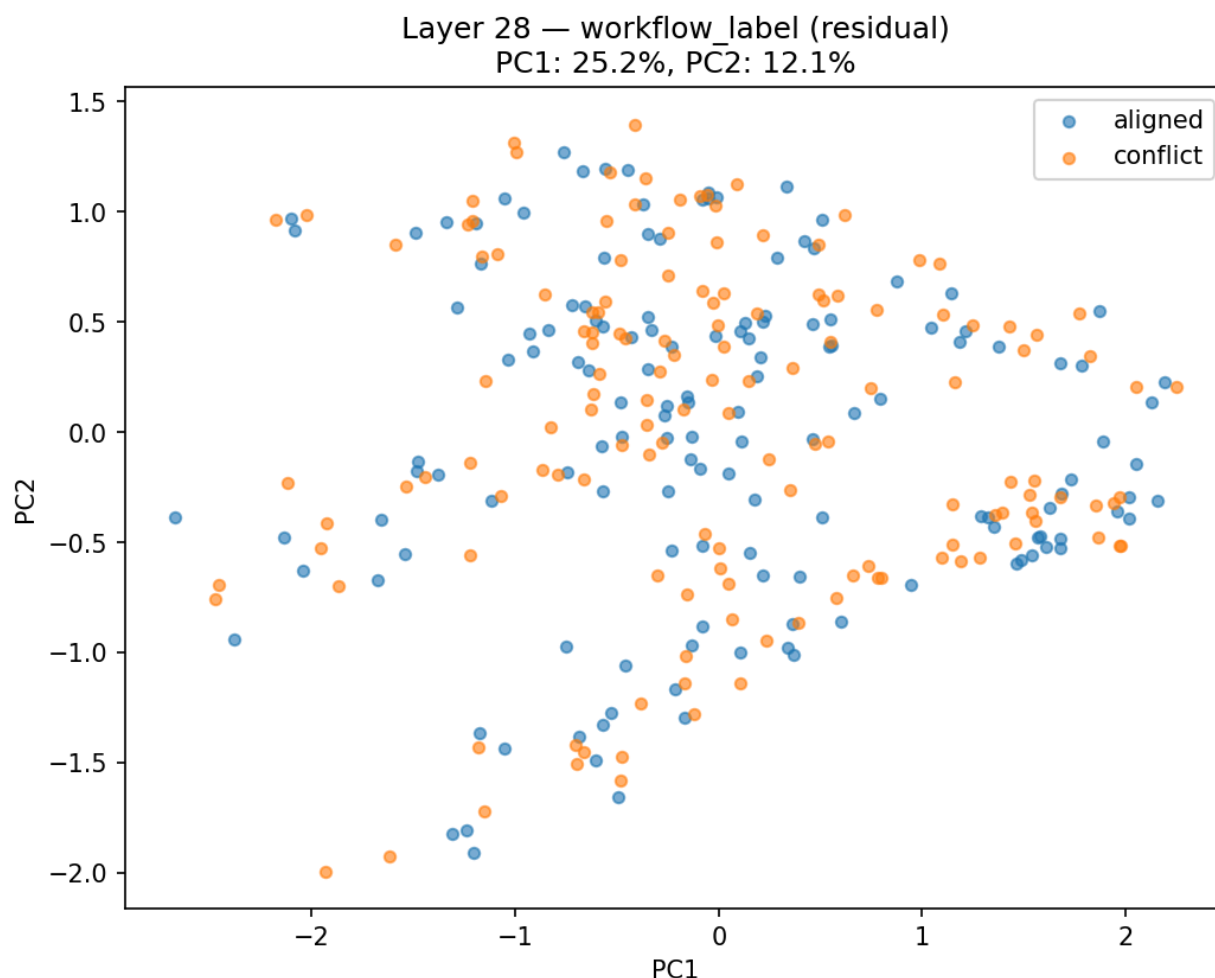


Figure 1: PCA of conflict vs. aligned residual activations at layer 28. PC1 and PC2 together explain 37% of variance.

Reading the PCA

The aligned and conflict points overlap heavily in this view — which might seem to contradict the 86% probe accuracy at this layer. It doesn't. PC1 and PC2 capture the directions of *maximum overall variance*, not the directions that best separate the labels. The residual stream is encoding many things at once (prompt content, family identity, token structure), and the conflict signal lives in a direction that is real but not dominant enough to show up as a top principal component.

The LDA projection (not shown) confirms this: when you project onto the *supervised* discriminant axis, aligned and conflict rows separate cleanly. The conflict direction is a learned feature of the representation, not the primary mode of variation.

2. Arbitration Signal Exists, But Is Confounded

On conflict-only rows, the residual stream carries information about *which side is winning* — but that information is heavily entangled with dataset structure.

The probe result

Stream	Best Layer	Balanced Accuracy	Std Dev
residual	20	0.712	0.062
residual	24	0.708	0.083
residual	44	0.700	0.079
router	16	0.648	0.069
router	28	0.647	0.087
router	44	0.622	0.117

The confound baseline

This is the key new result from Stage 2. Before interpreting the 71% probe as evidence of a latent arbitration mechanism, we need to ask: can we match that accuracy from dataset metadata alone?

Baseline	Balanced Accuracy	Std Dev	vs. Residual Probe (71.2%)
family + pressure	0.804	0.085	Above
family only	0.763	0.127	Above
user text n-gram	0.701	0.110	Tied
metadata (all)	0.674	0.054	Below
lexical IDs	0.651	0.103	Below
pressure only	0.573	0.033	Below
length / position	0.487	0.100	Below chance

KEY FINDING

Just knowing the conflict family and pressure bucket predicts the winner at 80.4% — materially better than the best residual probe (71.2%). The probe is not discovering information the dataset structure doesn't already carry.

Why this is interesting, not just a problem

The naive read is “the arbitration probe is confounded, so the result is weaker than we thought.” That’s true, but the confound structure itself is informative:

- **Family identity dominates.** The model resolves `activity_force_observe` conflicts differently from `trade_size_force_large` conflicts. This isn't noise — it means the model has learned family-specific resolution policies rather than a single abstract arbitration rule.
- **Pressure modulates within family.** Adding the pressure bucket on top of family pushes from 76% to 80%. The model's resolution is sensitive to *how hard* each section pushes, not just *what kind* of conflict it is.
- **Section length is irrelevant.** The `length_position` baseline is below chance (48.7%). Whatever drives the arbitration label, it is not a trivial surface feature. The confound is semantic, not mechanical.

So the story is: the model does not have one arbitration circuit. It has family-specific conflict resolution behaviors that are modulated by contextual pressure. That is a substantive finding about how LLMs handle instruction conflicts.

Visualizing the confound

The same PCA projection colored by arbitration label (left) versus by conflict family (right) tells the story immediately:

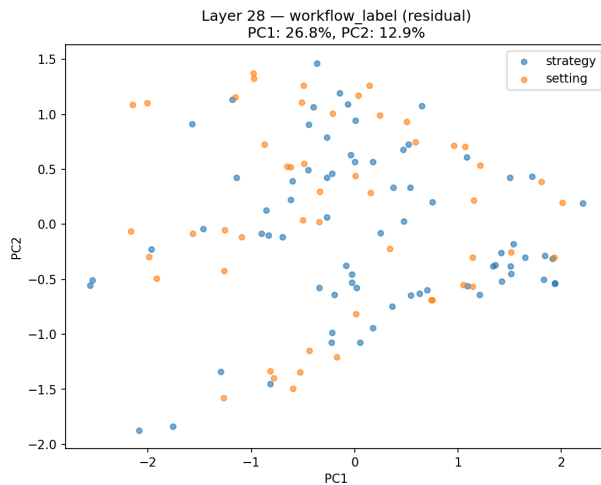


Figure 2: Layer 28 PCA colored by **arbitration label** (strategy vs. setting). No visible separation.

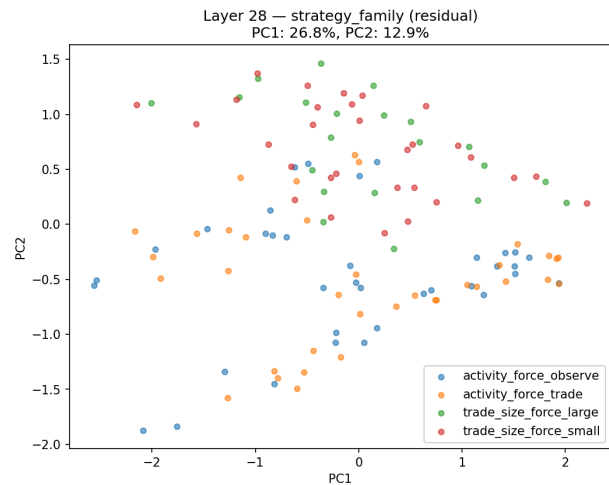


Figure 3: Same layer 28 PCA colored by **conflict family**. The structure maps onto families, not arbitration outcome.

The arbitration label doesn't organize the PCA space at all — but family identity does. Activity families (observe, trade) cluster on the left half of PC1; trade-size families (large, small) cluster on the right. This holds across layers and gets tighter through mid-layers.

The LDA projection makes the separation even more stark:

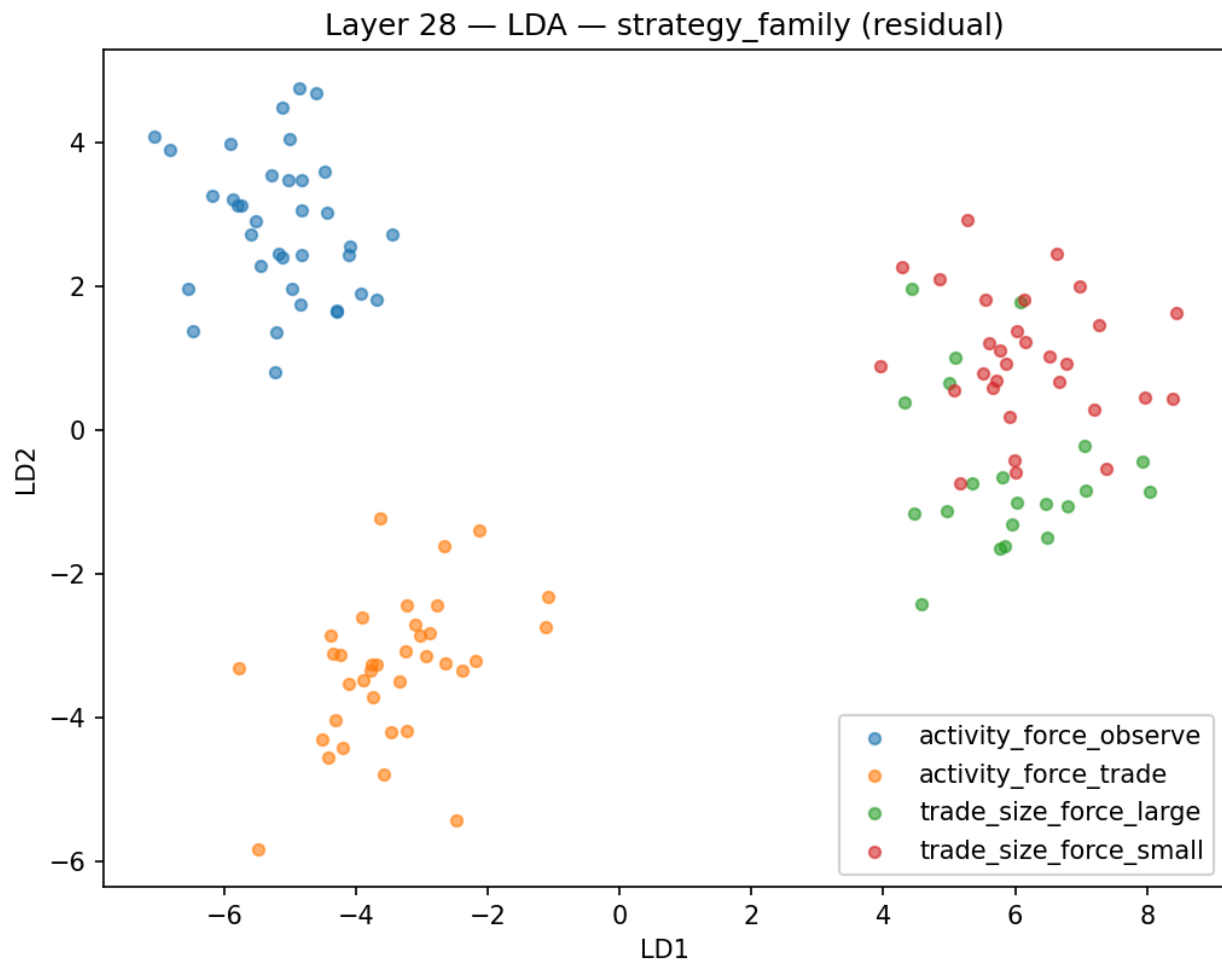


Figure 4: LDA projection at layer 28 colored by family. Near-perfect 4-way separation: LD1 splits activity vs. trade-size families; LD2 splits the specific variant within each pair. Almost zero overlap.

What this confirms

The dominant structure in the conflict-only residual stream is a 2×2 family organization: activity vs. trade-size on one axis, the specific conflict variant on the other. The centroid PCA puts 80.6% of between-family variance on PC1 alone — family identity is nearly one-dimensional.

An arbitration probe operating in this space is almost certainly riding family-correlated features rather than discovering a generic “which side wins” direction. The arbitration label cuts *across* the family clusters, which is why PCA doesn’t recover it. Any probe that achieves 71% without controlling for family could be doing so entirely through family-specific resolution patterns.

This does not invalidate the probe result — the probe is decoding real information about behavior. But it reframes what the probe is detecting: family-specific resolution policies, not an abstract arbitration mechanism. Within-family probes in Stage 3 will test whether anything remains after controlling for this structure.

3. Causal Intervention Is Behaviorally Active

Stage 1’s causal test was invalid (the model emitted <think> traces instead of JSON). Stage 2 fixed the harness, and the intervention now produces valid, directional effects.

Intervention target: SETTINGS mean @ layer 24, strength 1.0.

Condition	Strategy Rate	Setting Rate	Neither Rate
baseline	53.7%	45.5%	0.8%
setting_push	48.8%	48.8%	2.4%
strategy_push	62.6%	36.6%	0.8%

Key takeaway

The intervention works, but asymmetrically. strategy_push produces a clear +9pp shift toward strategy-following. setting_push is weaker (+3pp toward setting) and introduces some invalid outputs. This is consistent with the model having a slight strategy-default bias that is easier to amplify than to override.

Family-Level Causal Breakdown

The aggregate numbers hide important family-level structure.

Fam-ily	Base Strat.	Base Set.	Push→Strat.	Push→Set.	Strat. Delta	Set. Delta
activity_force_30	serve	6	31	5	+1	−1
activity_force_14	trade	22	16	23	+2	+1
trade_size_force_0	large	21	6	15	+6	0
trade_size_force_22	small	7	24	11	+2	+4

NOTABLE

The intervention bites hardest in trade_size_force_large, which is the family where the model most strongly defaults to setting-following behavior. Baseline has 0 strategy-favored rows; after strategy_push, 6 rows flip. The activity families barely move. This is further evidence that the causal patch is interacting with family-specific structure, not a generic arbitration direction.

4. Section Attribution

Where in the prompt does the arbitration signal come from? The section-attribution pass probed each prompt section independently on the conflict-only split.

Section	Pooling	Layer	Balanced Acc.	Note
PORTFOLIO	mean	28	0.792	Highest absolute, but confound risk.
MARKET	mean	24	0.784	Strong contextual-pressure confound.
SETTINGS	mean	24	0.765	Selected causal target.
SETTINGS	eos	36	0.764	Nearly tied.
STRATEGY	mean	36	0.743	Slightly weaker.

PORTFOLIO and MARKET score higher than SETTINGS, which is a confound warning: those sections carry contextual-pressure information that correlates with the arbitration label. SETTINGS was chosen as the intervention target because it is the cleanest *policy-source* section, even though it's not the strongest absolute readout.

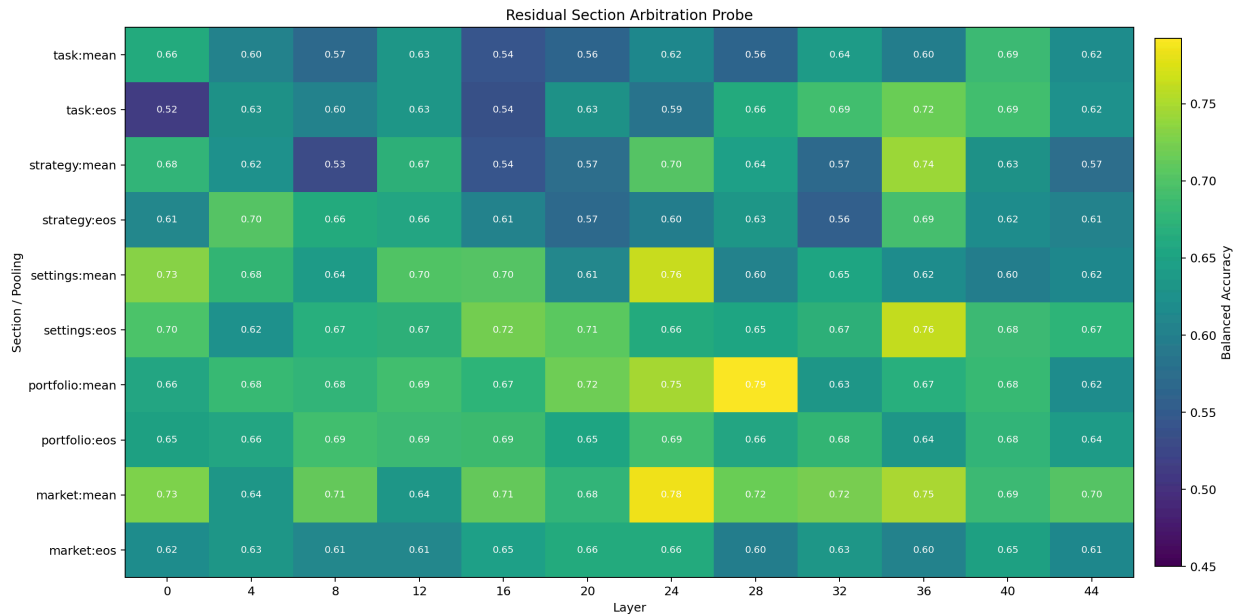


Figure 5: Grouped balanced accuracy by section, pooling, and layer for the conflict-only attribution pass.

5. Attention: Not Yet Informative

Section-aggregated attention mass barely distinguishes strategy-favored from setting-favored conflict resolution.

Layer	Anchor	Balanced Accuracy	Std Dev
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20	settings_eos	0.587	0.151
24	prompt_eos	0.554	0.030
36	strategy_eos	0.548	0.138
36	prompt_eos	0.544	0.036
24	settings_eos	0.535	0.169
20	prompt_eos	0.532	0.046

The best result (settings_eos at layer 20, 58.7%) has high variance and barely exceeds chance. The attention delta heatmap below confirms this: the mean deltas between strategy-favored and setting-favored rows are on the order of ± 0.003 , which is essentially zero at this sample size.

ompt-EOS Attention Delta (setting - strateg

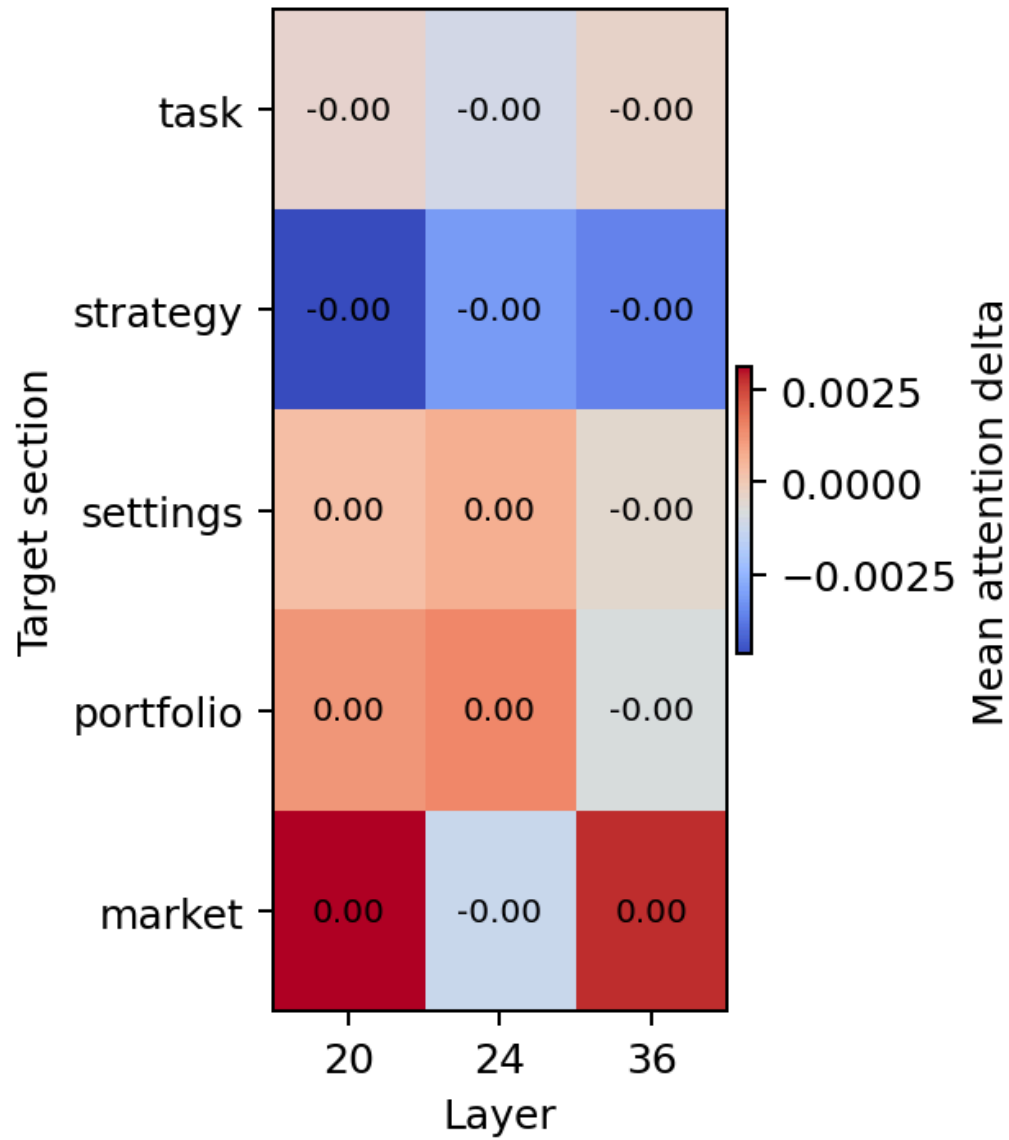


Figure 6: Mean attention delta (setting-favored minus strategy-favored) by target section and layer at the prompt EOS token. All values near zero.

What We Know Now

ESTABLISHED

- Conflict detection is strong and clean (92%)
- Causal intervention is behaviorally active
- Arbitration is family-specific, not generic

OPEN

- No clean cross-family arbitration mechanism yet

- Pressure modulates arbitration within families
- Section length is not a confound

- Causal effect is asymmetric and family-dependent
- PORTFOLIO / MARKET confound not isolated
- Attention is not informative at current resolution
- Router attribution still too sparse (N=25)

Interpretation: What Kind of Finding Is This?

The initial hypothesis was that the model might have a general-purpose arbitration circuit — a direction in the residual stream that encodes “follow source A over source B” regardless of conflict type. The evidence so far points to a different picture:

The model resolves conflicts categorically, not generically. When SETTINGS says “observe only” and STRATEGY says “trade aggressively,” the model resolves that conflict differently from when SETTINGS says “small positions” and STRATEGY says “go large.” Family identity alone predicts the winner at 76%. Adding contextual pressure gets to 80%.

This is not a failure of the experimental design. It is a real finding about how at least one model handles instruction conflicts: resolution is semantic and context-dependent, not a simple priority lookup. The practical implication is that prompting techniques that assume a stable “system prompt always wins” hierarchy are likely wrong — the actual hierarchy depends on *what kind* of instruction is being overridden and *how hard* the override pushes.

The next question is whether a residualized arbitration signal exists *after* controlling for family and pressure — i.e., whether there is also a generic component underneath the family-specific behavior. That is the main open question for Stage 3.

Next Steps

1. **Family-controlled arbitration probes.** Run within-family probes, or regress out family and pressure metadata before probing, to test whether a cross-family arbitration signal exists after confound control.
2. **Family-level causal evaluation.** Compare causal deltas within each family separately. The aggregate causal result mixes families where the patch does a lot (trade_size_force_large) with families where it does nothing (activity_force_observe).
3. **Isolate PORTFOLIO / MARKET confound.** These sections carry arbitration-adjacent signal that may just be pressure correlates. Test by shuffling or ablating them.
4. **Head-level attention.** Section-aggregated attention is too coarse. Move to per-head attention patterns conditioned on arbitration outcome.

5. **Increase router coverage.** Only 25 rows had usable full-span router data. Either fix the capture or extend the run.

Artifact Index

All paths relative to `projects/DX_TERMINAL/prompt_confusion/phase_04/outputs/`.

Stage 1

- `behavior_audit_16474bceae4e.json`
- `analysis_full_prompt_eos_probe/62fc89fd861b/results.json`
- `analysis_full_prompt_eos_pca/f4c70fcbac6d/`
- `analysis_conflict_readout_residual/6924c2b36f43/results.json`
- `analysis_conflict_readout_pca/331de685cbc1/`
- `analysis_conflict_readout_router/4aeb882aea9d/results.json`
- `conflict_arbitration_stage1/section_attribution/`
- `conflict_arbitration_stage1/causal_check/` (invalid — harness bug)

Stage 2

- `analysis_conflict_readout_pca_by_family/` (family-colored PCA/LDA/centroid plots)
- `conflict_arbitration_stage2/causal_check/summary.json`
- `conflict_arbitration_stage2/causal_check/row_level.parquet`
- `conflict_arbitration_stage2/confound_checks/summary.json`
- `conflict_arbitration_stage2/confound_checks/baseline_results.parquet`
- `conflict_arbitration_stage2/attention_summary/summary.json`
- `conflict_arbitration_stage2/attention_summary/row_level.parquet`
- `conflict_arbitration_stage2/attention_summary/prompt_eos_attention_delta.png`

Caveats, Limitations, and Isolation Strategy

This appendix collects the methodological cautions that should inform how the main results are read and what Stage 3 needs to address.

A. How Much Should We Trust the Family Separation?

The LDA plot at layer 28 shows near-perfect 4-way family separation. That is visually striking but should be interpreted carefully.

LDA is designed to separate classes

LDA finds the projection that **maximally separates** the groups. In a 2048-dimensional residual stream with only 30 points per family, LDA has enormous freedom to find separating directions. It would be more surprising if it *couldn't* find clean separation. The near-zero overlap in the LDA is partly the method flattering the data.

The unsupervised PCA tells a more honest story. Family structure is visible in PCA too — activity families cluster on the left of PC1, trade-size families on the right — but with substantial overlap, especially at later layers. The top two principal components explain only 40% of variance, so most of the residual stream is encoding things unrelated to family identity.

What makes the family finding more than a methodological artifact:

- The **confound baselines** confirm it statistically, independent of any projection method. `family_only` predicts the arbitration label at 76.3% using a simple classifier on a categorical variable — no high-dimensional geometry involved.
- The **PCA** shows family structure in an unsupervised setting, before any label information is used.
- The **causal intervention** bites differently across families, which would not happen if families were just label noise.

What does **not** follow:

- That the model has “dedicated family circuits.” The families have genuinely different prompt content — different vocabulary, different section phrasing. Some separation is trivially expected because the model is encoding “these prompts contain different words.”
- That 76% family-baseline accuracy means the probe is doing nothing beyond family. It could be capturing a mix of family features and genuine within-family arbitration signal. We cannot distinguish these yet.

B. Sample Size Constraints

Analysis	N	Concern
Conflict detection probe	288	Adequate for grouped 5-fold.
Arbitration probe	123	Tight. 5-fold grouped CV on 43 groups.
Per-family arbitration	~30	Too small for reliable within-family probes.
Router section attribution	25	Not interpretable.
Causal check (per family)	~30	Directional only; single-row flips matter.

The per-family N of 30 is the binding constraint for Stage 3. Within-family probing with 5-fold CV on 30 examples will have high variance. Options:

- **Expand the dataset.** Add more conflict variants per family, or add new families. This is the cleanest fix but requires new data generation and capture.
- **Pool across families with family as a covariate.** Regress out family (and pressure) from the activations before probing. This preserves N=123 while controlling for the confound, but assumes the confound is additive.
- **Permutation-based significance.** For small N, replace accuracy with a permutation test to get proper p-values rather than relying on point estimates.

C. What the Causal Result Does and Does Not Show

The causal intervention moves behavior in the expected direction, but interpretation is limited:

- **Asymmetry.** `strategy_push` produces +9pp shift; `setting_push` produces only +3pp. This could mean the direction is not symmetric, or that the model has a strategy-default bias that is easier to amplify than override.
- **Family dependence.** The intervention bites hardest in `trade_size_force_large` (6 row flips from 0 baseline) and barely moves the activity families. If the intervention were acting on a generic arbitration direction, we would expect it to move all families. Instead it interacts with family-specific structure.
- **Strength = 1.0.** We used a single intervention strength. A strength sweep would reveal whether the effect is linear, saturating, or has a threshold — important for distinguishing a genuine direction from a perturbation artifact.
- **No same-label control.** We did not run the intervention on **aligned** rows to check whether it disrupts behavior when no conflict exists. A clean causal direction should have no effect on aligned prompts.

D. Isolation Strategy for Stage 3

The main open question is: **does a cross-family arbitration signal exist after controlling for family and pressure?** The following analyses would address this:

Priority	Analysis	What it would show
1	Residualize activations: regress out family + pressure, then re-probe	Whether a generic arbitration component exists underneath family-specific behavior
2	Within-family probes (where N allows)	Whether arbitration signal exists within a single conflict type
3	Causal intervention on aligned rows (same-label control)	Whether the intervention direction is specific to conflict or a general perturbation
4	Causal strength sweep (0.25, 0.5, 1.0, 2.0)	Whether the effect is linear and dose-dependent
5	Expand dataset to 60+ rows per family	Statistical power for within-family analyses
6	Cross-family generalization: train probe on 3 families, test on held-out 4th	Whether the probe learns anything that transfers across conflict types

Item 6 is the sharpest test. If a probe trained on activity families can predict arbitration outcome on trade-size families (or vice versa), that is strong evidence for a generic signal. If it cannot, the arbitration mechanism is family-specific all the way down.

Prompt Confusion Phase 04 Combined Checkpoint — stages 1 and 2. Generated from local outputs under phase_04/outputs. Behavior run 16474bce, conflict probe 62fc89fd, arbitration probes 6924c2b3 / 4aeb882e, PCA runs f4c70fca / 331de685, family PCA from analysis_conflict_readout_pca_by_family, section attribution and causal from conflict_arbitration_stage1 and conflict_arbitration_stage2.